Multiple Choice Questions

Each question is worth 1 mark. No partial credit.

IMPORTANT NOTES:

1. The questions with numerical answers are intended to help you practice applying formulas. You will not be required to calculate numerical solutions on the test. Please refer to Term Test 1 solutions and cover page for the format.
2. You will not be required to memorize formulas. These questions are intended to help you recognize which formulas apply in different situations. You should complete these using the aid sheet on the Term test 2 page.
3. These questions were generated with the help of AI. To create them, I uploaded my slides to Gemini and asked it to generate questions aligned with the learning outcomes. As a result, some questions may be quite specific. However, I think they are still a great way to revisit and practice the material, even if you complete them open book.
4. I skimmed over these questions but if you see anything suspicious please don't hesitate to message me on piazza.
5. Generative AI is used to create first drafts of some multiple choice questions on the test. The difference between these practice questions and the actual test questions is that I refine the test questions afterward, including revising wording, improving distractors, and removing options that are obviously incorrect (in a silly way).

OVERVIEW OF THIS DOCUMENT:

1. Multiple choice questions, separated by lecture
2. Answer key, separated by lecture
3. High level concept summaries for lectures 5-7

Lecture 5

Q1. An investigator wants to estimate the total annual timber yield across a dense managed forest zone using ratio estimation. She has access to historical baseline maps providing a completely accurate, known population total for canopy density ($t_x > 0$). Which of the following parameters must be tracked or true for the final ratio estimator of the population total ($\hat{t}_{yr}$) to be mathematically well-defined?

(A) The target variable y must have a strictly negative correlation with the auxiliary variable x.

(B) The total number of individual sampled plots n must be exactly equal to the total population size N.

(C) The sample sum of the auxiliary variable ($\sum_{i \in S} x_i$) must be strictly greater than zero, ensuring $\overline{x}_S \neq 0$.

(D) The population variance of the target variable (S_y^2) must asymptotically approach infinity.

(E) The vertical intercept of the underlying linear trend line must be strictly non-zero ($B_0 \neq 0$).

Q2. Under a standard simple random sampling design without replacement, an analyst estimates a ratio using the classical parameter formula $\hat{B} = \frac{\overline{y}_S}{\overline{x}_S}$. Which of the following expressions provides the correct mathematical structural definition for the estimated sampling variance $\hat{V}(\hat{B})$?

(A) $\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n}$

(B) $\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n\overline{x}_S^2}$

(C) $\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \left(\frac{\overline{x}_U}{\overline{x}_S}\right)^2 \frac{s_e^2}{n}$

(D) $\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \left(\frac{t_x}{\overline{x}_S}\right)^2 \frac{s_e^2}{n}$

(E) $\hat{V}(\hat{B}) = \left(1 - \frac{n}{N}\right) \frac{N^2 s_e^2}{n}$

Q3. A regional utility service uses simple random sampling to estimate the average household electricity bills (y) in a city, using residential square footage (x) as a tracking auxiliary variable. The true population mean square footage is known from municipal records to be $\overline{x}_U = 1500$. A sample of households reveals a sample mean electricity bill of $\overline{y}_S = 820$ and a sample mean square footage of $\overline{x}_S = 1600$. Compute the ratio estimate of the population mean electricity usage ($\hat{\overline{y}}_r$).

(A) 874.67

(B) 768.75

(C) 820.00

(D) 843.15

(E) 795.30

Q4. Referring to the exact scenario described in the previous question, determine whether the ratio adjustment increased or decreased the ordinary unadjusted sample mean estimate ($\overline{y}_S$), and select the correct intuitive statistical justification for this shift.

(A) It increased the estimate because the sampled households were smaller on average than the city population.

(B) It decreased the estimate because the sampled households were larger on average ($\overline{x}_S = 1600$) than the general population ($\overline{x}_U = 1500$), indicating that the raw sample mean was likely an overestimate.

(C) It left the estimate unchanged because calibration adjustments only impact population total metrics.

(D) It decreased the estimate because the target variable and auxiliary variable have a negative correlation coefficient.

(E) It increased the estimate because the proportionality factor adjustments always shift estimates upward.

Q5. An environmental researcher selects an SRS of $n = 40$ ground plots from a finite conservation territory of $N = 400$ plots to estimate total vegetation biomass (y) using canopy cover (x) as an auxiliary variable. The true population total canopy cover is known to be $t_x = 12,000$ m^2. If the cumulative sample biomass is $\sum_{i\in S} y_i = 14,800$ kg and the cumulative sample canopy cover is $\sum_{i\in S} x_i = 1,184$ m^2, calculate the calibrated combined ratio estimate of the population total biomass ($\hat{t}_{yr}$).

(A) 148,000 kg

(B) 150,000 kg

(C) 118,400 kg

(D) 125,000 kg

(E) 132,500 kg

Q6. Using the vegetation data from the previous question, the sample residual variance is found to be $s_e^2 = 16.5$. Compute the estimated sampling variance $\hat{V}(\hat{t}_{yr})$ of your population total estimator.

(A) 13,500.00

(B) 59352.17

(C) 33,412.50

(D) 41,250.00

(E) 3,712.50

Q7. When analyzing why a ratio estimator can outperform an ordinary simple random sample mean, we evaluate the mean squared error comparison. Under which of the following conditions is the MSE of a ratio estimator guaranteed to be smaller than the variance of the unadjusted sample mean estimator?

(A) $R \times \frac{CV(x)}{CV(y)} > \frac{1}{2}$

(B) $R \times \frac{CV(y)}{CV(x)} > \frac{1}{2}$

(C) $R \times \frac{CV(y)}{CV(x)} < \frac{1}{2}$

(D) $R = 0$ identically

(E) $\frac{CV(y)}{CV(x)} \approx 0$

Q8. An auditor takes an SRS of $n = 50$ items from an asset framework of $N = 1000$ elements to inspect true accounting values (y) against values reported in a ledger (x). In this sample, $\overline{y}_S = 98$, $\overline{x}_S = 102$, the estimated residual variance is $s_e^2 = 25$, and the true population mean of reported values is known to be $\overline{x}_U = 100$. Calculate the estimated ratio estimator of the population mean $\hat{\overline{y}}_r$.

(A) 98.00

(B) 102.00

(C) 96.08

(D) 100.00

(E) 94.12

Q9. Let s_e^2 be the sample residual variance used across ratio adjustment frameworks. Which of the following expressions provides the correct computational structure for calculating s_e^2 from sample items under a standard ratio model?

(A) $s_e^2 = \frac{1}{n-1}\sum_{i\in S}(y_i - \overline{y}_S)^2$

(B) $s_e^2 = \frac{1}{n-1}\sum_{i\in S}(x_i - \overline{x}_S)^2$

(C) $s_e^2 = \frac{1}{n-1}\sum_{i\in S}(y_i - \hat{B}x_i)^2$

(D) $s_e^2 = \frac{1}{n-1}\sum_{i\in S}(y_i - \hat{B}_0 - \hat{B}_1 x_i)^2$

(E) $s_e^2 = \frac{1}{n-1}\sum_{i\in S}(y_i - \overline{y}_S - \hat{B}x_i)^2$

Q10. A statistician notes that a dataset has a very small sample residual variance s_e^2. What does this indicate about the underlying structure of the data elements?

(A) There are extreme, unpredictable deviations from the straight-line regression model.

(B) The variables x and y are completely independent and exhibit a correlation close to zero.

(C) There are minimal deviations from the trend line, meaning that $\hat{B}x_i$ serves as a highly accurate predictor for y_i.

(D) The coefficient of variation of the denominator variable is drastically inflated.

(E) The finite population correction factor will converge to 1, inflating the standard error.

Q11. A foundational lecture slide establishes that the approximate bias of a ratio estimator can be tracked using specific population parameters. For both the ratio slope estimator $\hat{B}$ and the population mean ratio estimator $\hat{\overline{y}}_r$, which structural term maps the common core component inside the bias equations?

(A) $(BS_x^2 - RS_xS_y)$

(B) $(B^2S_x^2 + S_y^2)$

(C) $(S_y^2 - B^2S_x^2)$

(D) $(RS_xS_y - BS_x^2)$

(E) $(S_{xy} - S_x^2)$

Q12. An analyst must choose between a ratio estimator ($\hat{\bar{y}}_r$) and a linear regression estimator ($\hat{\bar{y}}_{reg}$). What is the core structural difference between the modeling assumptions of these two adjustment techniques?

(A) Regression estimation assumes a perfectly proportional relationship passing through the origin, whereas ratio estimation does not.

(B) Ratio estimation assumes the relationship between y and x goes approximately through the origin ($y \approx Bx$), whereas regression estimation allows for a strictly non-zero vertical intercept ($y \approx B_0 + B_1 x$).

(C) Regression estimation requires knowing the population variance S_x^2, while ratio estimation ignores all auxiliary totals.

(D) Ratio estimation is strictly unbiased for all sample sizes, whereas linear regression exhibits massive negative bias.

(E) Regression estimation can only handle negative correlations, whereas ratio estimation requires positive correlation.

Q13. A simple random sample of $n = 5$ students is drawn from a class population of $N = 100$ to estimate exam performance (y) using hours studied (x) as an auxiliary variable. The known population mean study time is 6 hours. The sample data are: Student 1 $(2, 58)$, Student 2 $(4, 65)$, Student 3 $(5, 70)$, Student 4 $(7, 78)$, Student 5 $(8, 85)$. If the calculated sample averages are $\overline{x}_S = 5.2$ and $\overline{y}_S = 71.2$, and the estimated slope is $\hat{B}_1 = 4.3836$, compute the regression estimate of the population mean score ($\hat{\bar{y}}_{reg}$).

(A) 71.20

(B) 74.71

(C) 67.69

(D) 75.58

(E) 73.12

Q14. Reviewing the exact regression calibration equation $\hat{\bar{y}}_{reg} = \overline{y}_S + \hat{B}_1(\overline{x}_U - \overline{x}_S)$, suppose the sample mean of x is strictly less than the known population mean ($\overline{x}_S < \overline{x}_U$) and the estimated slope is positive ($\hat{B}_1 > 0$). What direction of adjustment is executed on the unadjusted sample mean ($\overline{y}_S$)?

(A) It adjusts the estimate downward because the difference ($\overline{x}_U - \overline{x}_S$) is negative.

(B) It scales the estimate by a multiplicative factor tracking the proportion $\overline{x}_U / \overline{x}_S$.

(C) It adjusts the estimate upward because the sample underestimated the true auxiliary average, meaning the raw sample target mean is likely lower than the true population mean.

(D) It leaves the sample mean completely unadjusted because regression estimation is strictly distribution-free.

(E) It shifts the estimate downward to correct for potential high leverage items.

Q15. Under a standard simple random sampling design tracking linear regression calibrations, what are the approximate biases of the regression total estimator $\hat{t}_{yreg}$ and the regression mean estimator $\hat{\overline{y}}_{reg}$?

(A) Both estimators exhibit an approximate bias of 0.

(B) The total estimator is biased by $(1 - n/N)$, while the mean estimator is unbiased.

(C) Both equations exhibit a substantial negative bias tracking the value of the non-zero intercept.

(D) The approximate bias is equal to the sample residual variance divided by n.

(E) The bias is structurally identical to the ratio equation bias $(BS_x^2 - RS_xS_y)$.

Q16. Let s_y^2 be the unadjusted sample variance of the target variable and let r represent the sample correlation coefficient. Which of the following defines the correct formula for the estimated sampling variance $\hat{V}(\hat{\overline{y}}_{reg})$ of the regression mean estimator?

(A) $\hat{V}(\hat{\overline{y}}_{reg}) = \left(1 - \frac{n}{N}\right) \frac{1}{n} s_y^2(1 - r^2)$

(B) $\hat{V}(\hat{\overline{y}}_{reg}) = \left(1 - \frac{n}{N}\right) \frac{N^2}{n} s_y^2(1 - r^2)$

(C) $\hat{V}(\hat{\overline{y}}_{reg}) = \left(1 - \frac{n}{N}\right) \frac{s_e^2}{n\overline{x}_S^2}$

(D) $\hat{V}(\hat{\overline{y}}_{reg}) = \left(1 - \frac{n}{N}\right) \sum_{h=1}^{H} \frac{N_h^2}{N^2} \frac{s_h^2}{n}$

(E) $\hat{V}(\hat{\overline{y}}_{reg}) = \left(1 - \frac{n}{N}\right) \frac{s_y^2}{n}(1 + r^2)$

Q17. When constructing standard confidence intervals for parameters estimated via linear regression estimation models, what is the exact number of degrees of freedom used for selecting the critical value multiplier from a Student's t-distribution?

(A) $n - 1$

(B) $n - 2$

(C) $n_h - 1$

(D) $N - 1$

(E) n

Q18. A research study explores the baseline calibration properties of regression estimation. Which of the following statements is strictly TRUE regarding the behavior of the regression mean estimator $(\hat{\overline{y}}_{reg})$?

(A) Regression estimation only improves precision when the true sample correlation coefficient r is strictly positive.

(B) If the drawn sample happens to be perfectly representative of the auxiliary variable such that $\overline{x}_S = \overline{x}_U$, then the regression estimator simplifies exactly to the ordinary sample mean $\overline{y}_S$.

(C) Regression estimation assumes that the absolute relationship between y and x is deterministic and free of any residual errors.

(D) The estimator performs best when the correlation coefficient between x and y approaches exactly zero.

(E) The vertical adjustment factor requires a multi-stage cluster frame to select initial slopes.

Q19. The coefficient of variation is a critical diagnostic parameter defined across survey analysis workflows. Suppose Dataset A has a population mean of 100 and a standard deviation of 10, while Dataset B has a population mean of 5 and a standard deviation of 2. Which dataset has a higher coefficient of variation, and what does this signify?

(A) Dataset A, because its absolute standard deviation is larger.

(B) Dataset B, because its relative variability around its mean ($CV = 0.40$) is larger than Dataset A's relative variability ($CV = 0.10$).

(C) Both datasets have identical coefficients of variation because they scale linearly.

(D) Dataset B, because its absolute standard deviation is smaller relative to the population size.

(E) Dataset A, because a high mean automatically forces an inflated variance tracking curve.

Q20. An investigator wants to construct a 95% confidence interval for an asset ratio estimate. She calculates the ratio slope as $\hat{B} = 0.9608$, the sample size as $n = 50$, and the estimated variance of the ratio as $\hat{V}(\hat{B}) = 0.0000456$. Assuming the critical multiplier from the t-distribution is approximately $t_{\alpha/2,49} = 2.0096$, compute the resulting 95% confidence interval.

(A) $[0.9472, 0.9744]$

(B) $[0.9517, 0.9699]$

(C) $[0.9325, 0.9891]$

(D) $[0.8950, 1.0266]$

(E) $[0.9142, 1.0074]$

Q21. An environmental consultant sets up a directional hypothesis test to check if total forest biomass has significantly diverged from a historical baseline total of 150,000 kg. What is the correct statistical structure for the null and alternative hypotheses to evaluate this research question?

(A) $H_0 : t_y = 150,000$ vs $H_A : t_y \neq 150,000$

(B) $H_0 : \hat{t}_{yr} \leq 150,000$ vs $H_A : \hat{t}_{yr} > 150,000$

(C) $H_0 : t_y \geq 150,000$ vs $H_A : t_y < 150,000$

(D) $H_0 : \hat{B} = 1.00$ vs $H_A : \hat{B} \neq 1.00$

(E) $H_0 : \overline{y}_U = 150,000$ vs $H_A : \overline{y}_U \neq 150,000$

Q22. A colleague creates two distinct theoretical simulation frameworks to evaluate the performance of ratio estimation models:

- Scenario A: The relationship between y and x is perfectly linear, passing exactly through the origin, but the variance of x in the population is exceptionally large ($S_x^2 \to \infty$).
- Scenario B: The relationship between y and x is perfectly linear but exhibits a strictly non-zero vertical intercept.

Evaluate the behavior of the approximate bias of the ratio total estimator $\hat{t}_{yr}$ under Scenario A vs Scenario B.

(A) Scenario A will cause the bias to explode to infinity, while Scenario B results in zero bias.

(B) Scenario A will exhibit zero approximate bias because the proportionality assumption holds perfectly ($y = Bx$), whereas Scenario B will exhibit non-zero bias due to the intercept violation.

(C) Both simulation settings will yield an identical approximate bias of zero.

(D) Scenario B will exhibit zero bias because linear alignments protect against intercept shifts.

(E) Scenario A will yield a large negative bias tracking the size of the finite population correction factor.

Q23. Referring to the exact same simulation parameters described in the previous question, evaluate the behavior of the variance of the ratio total estimator $\hat{t}_{yr}$ under Scenario A.

(A) The variance will converge exactly to zero because the linear relationship is perfect.

(B) The variance will explode towards infinity ($V(\hat{t}_{yr}) \to \infty$) because the variance component equations are scaled by the exceptionally large population variance of the auxiliary variable (S_x^2).

(C) The variance will remain bounded by the standard simple random sampling target variance s_y^2.

(D) The variance will depend exclusively on the sample correlation coefficient r.

(E) The variance becomes mathematically undefined because denominators cancel out.

Q24. Let $CV(y)$ be the coefficient of variation of the target variable and $CV(x)$ be the coefficient of variation of the auxiliary variable. If a diagnostic review shows that $CV(y)/CV(x) \approx 1$, how should this relationship be interpreted structurally?

(A) Variable y varies similarly relative to its mean than variable x does relative to its mean.

(B) The absolute variance of variable y is identical to the absolute variance of variable x.

(C) The population correlation coefficient between the variables must be equal to 0.5.

(D) Ratio estimation is mathematically guaranteed to be less efficient than an unadjusted SRS.

(E) The regression slope through the origin will equal exactly 1.00.

Q25. An investigator wants to explain to a client why ratio estimation calibrates or increases precision in large-scale operations. Which of the following provides the most comprehensive conceptual explanation?

(A) Ratio estimation increases precision by forcing all sample metrics to follow a standard normal distribution.

(B) It works best when the auxiliary variable x tracks y closely (strong correlation) and carries useful information about how y shifts relative to its mean, allowing the model to adjust for sample imbalances.

(C) It guarantees that the finite population correction factor drops to zero, neutralizing sampling error.

(D) It completely removes the need to collect target data inputs, relying entirely on known census totals.

(E) It fits a complex polynomial regression curve that minimizes all potential multi-stage clustering effects.

Lecture 6

Q1. A researcher uses a ratio estimator $\hat{\bar{y}}_r = \hat{B}\bar{x}_U$ to adjust simple random sample estimates. Under which of the following conditions is the approximate bias of this ratio estimator guaranteed to be minimized?

(A) When the auxiliary variable x and target variable y have an exceptionally high negative correlation.

(B) When the sample size n is small relative to the population size N.

(C) When the true regression relationship between y and x passes exactly through the origin ($y \approx Bx$).

(D) When the coefficient of variation of x is substantially larger than the coefficient of variation of y.

(E) When the sampling fraction n/N approaches zero.

Q2. An auditor takes a simple random sample of $n = 50$ items from an inventory population of $N = 1000$ to estimate the ratio of actual book value (y) to reported book value (x). The sample mean of actual values is $\bar{y}_S = \$98$, the sample mean of reported values is $\bar{x}_S = \$102$, and the estimated residual variance is $s_e^2 = 25$. If the population mean of reported values is known to be $\bar{x}_U = \$100$, calculate the estimated ratio $\hat{B}$ and its estimated sampling variance $\hat{V}(\hat{B})$.

(A) $\hat{B} = 0.9608$, $\hat{V}(\hat{B}) = 0.00456$

(B) $\hat{B} = 1.0408$, $\hat{V}(\hat{B}) = 0.00048$

(C) $\hat{B} = 0.9608$, $\hat{V}(\hat{B}) = 0.000456$

(D) $\hat{B} = 0.9608$, $\hat{V}(\hat{B}) = 0.0000456$

(E) $\hat{B} = 0.9800$, $\hat{V}(\hat{B}) = 0.000500$

Q3. A simple random sample of $n = 5$ farms is selected from a finite population of $N = 100$ farms to estimate average crop yield (y) using farm size (x) as an auxiliary variable. You are given that $\bar{x}_U = 48$. The sample values are: Farm 1 ($x = 40, y = 52$), Farm 2 ($x = 55, y = 70$), Farm 3 ($x = 60, y = 74$), Farm 4 ($x = 45, y = 58$), Farm 5 ($x = 50, y = 63$). Compute the ratio estimate of the population mean crop yield $\hat{\bar{y}}_r$.

(A) 63.40 tons

(B) 60.86 tons

(C) 66.04 tons

(D) 50.00 tons

(E) 58.32 tons

Q4. When framing ratio estimation using weight adjustments, we define an adjusted sampling weight $\omega_i^* = g_i\omega_i$. Which statement correctly describes the nature of the calibration factor g_i under a simple random sampling design?

(A) g_i is a fixed population constant equal to N/n.

(B) $g_i = t_x/\hat{t}_x$, making it a random variable that varies from sample to sample.

(C) g_i is forced to be equal to 1 if x and y are negatively correlated.

(D) $g_i = \hat{t}_y/t_x$, calibrating the target total directly.

(E) g_i depends exclusively on the cluster sizes M_i within the design.

Q5. What is the critical distinction between Stratified Sampling and Domain Estimation regarding sample size?

(A) In stratified sampling, the sample sizes are random variables, whereas in domain estimation they are fixed by design.

(B) In stratified sampling, sample sizes are fixed and pre-determined; in domain estimation, sample sizes within sub-populations are random variables.

(C) Both designs maintain deterministic, strictly fixed sample sizes across all sub-populations.

(D) Domain estimation requires a census of every subpopulation, while stratified sampling does not.

(E) Stratified sampling can only be used with continuous auxiliary data, while domain estimation requires binary indicators.

Q6. Suppose a researcher wants to show that the domain mean estimator $\overline{y}_d = \frac{1}{n_d} \sum_{i \in S_d} y_i$ is a specific form of a ratio estimator $\hat{B} = \overline{u}_S/$
$overlinex_S$. If we define an indicator variable $x_i = 1$ when unit $i \in \mathcal{U}_d$ and 0 otherwise, what must the transformed variable u_i be defined as?

(A) $u_i = y_i$ for all units in the population.

(B) $u_i = 1/y_i$ if $i \in \mathcal{U}_d$ and 0 otherwise.

(C) $u_i = y_i$ if $i \in \mathcal{U}_d$ and 0 if $i \notin \mathcal{U}_d$.

(D) $u_i = y_i - \overline{y}_S$ for all items.

(E) $u_i = n_d/n$ for all sampled components.

Q7. A survey is conducted using a simple random sample of size $n = 100$. Post-stratification is executed by age category. The true population counts are $N_1 = 6000$ individuals under 40, and $N_2 = 4000$ individuals aged 40 and over. The sample results show: Under 40 ($n_1 = 70$, $\overline{y}_{1S} = \$48,000$) and 40 and over ($n_2 = 30$, $\overline{y}_{2S} = \$62,000$). Compute the post-stratified estimate of the population mean income $\overline{y}_{post}$.

(A) \$52,200

(B) \$55,000

(C) \$53,600

(D) \$50,800

(E) \$51,400

Q8. In which practical scenario is Post-Stratification most appropriate over standard Stratified Random Sampling?

(A) When the sampling frame completely details the stratum membership of every unit before the sample is drawn.

(B) When cluster sizes are small and homogeneous.

(C) When the auxiliary variable is highly relevant to the variable of interest, but this information is completely missing from the sampling frame during the selection phase.

(D) When the relationship between x and y is strictly non-linear and passes through the origin.

(E) When you want to increase the variance of the estimator to make confidence intervals wider.

Q9. For a stratified sampling design, the combined ratio estimator for a population total is given by $\hat{t}_{yrc} = \hat{B}t_x$, where $\hat{B} = \hat{t}_{y,str}/\hat{t}_{x,str}$. Under proportional allocation ($n_h/n = N_h/N$), show what the structural form of $\hat{B}$ simplifies to.

(A) $\hat{B} = \frac{\sum_{h=1}^{H} N_h \overline{y}_{hS}}{\sum_{h=1}^{H} N_h \overline{x}_{hS}}$

(B) $\hat{B} = \frac{\sum_{h=1}^{H} \sum_{i \in S_h} y_{hi}}{\sum_{h=1}^{H} \sum_{i \in S_h} x_{hi}}$

(C) $\hat{B} = \sum_{h=1}^{H} \frac{\overline{y}_{hS}}{\overline{x}_{hS}}$

(D) $\hat{B} = \frac{\overline{y}_{1S} + overliney_{2S}}{\overline{x}_{1S} + \overline{x}_{2S}}$

(E) $\hat{B} = 1.00$ identically across all structural iterations.

Q10. To achieve optimal statistical precision when designing a survey, what are the ideal structural characteristics inside strata versus inside clusters?

(A) Elements within a stratum should be heterogeneous; elements within a cluster should be homogeneous.

(B) Both strata and clusters should contain highly homogeneous elements to minimize variance.

(C) Elements within each stratum should be homogeneous (similar values), whereas elements within each cluster should be heterogeneous (diverse values).

(D) Both strata and clusters should be perfectly representative, identical copies of the full population frame.

(E) Strata must rely exclusively on geographic boundaries, while clusters must ignore design weights entirely.

Q11. A researcher is comparing ratio estimation and linear regression estimation for a specific survey dataset. Under which structural population scenario will the regression estimator ($\hat{\overline{y}}_{reg}$) yield a significantly more precise estimate (lower MSE) than the standard ratio estimator ($\hat{\overline{y}}_{r}$)?

(A) When the relationship between the target variable y and the auxiliary variable x is perfectly linear and passes directly through the origin $(0, 0)$.

(B) When the true relationship between y and x is linear but exhibits a strictly large, non-zero vertical intercept ($B_0 \neq 0$).

(C) When the population correlation coefficient R between the two variables is exactly equal to zero.

(D) When the relative variability of the auxiliary variable is vastly superior to the relative variability of the target variable.

(E) When the finite population correction factor approaches zero, neutralizing the impact of the residual variance.

Q12. Suppose you are deriving the sampling variance of the domain mean estimator $\overline{y}_d$ by treating it as a ratio of two sample means $(\overline{u}_S/\overline{x}_S)$. If $s^2_{yd} = \frac{1}{n_d-1}\sum_{i\in S_d}(y_i - \overline{y}_d)^2$ represents the sample variance within domain d, which of the following expressions represents the mathematically sound, unsimplified sample residual variance s^2_e used in the standard ratio variance formula?

(A) $s^2_e = \frac{1}{n-1}\sum_{i\in S}(y_i - \overline{y}_S)^2$

(B) $s^2_e = \frac{1}{n-1}\sum_{i\in S_d}(y_i - \overline{y}_d x_i)^2$

(C) $s^2_e = \frac{1}{n-1}\sum_{i\in S}(u_i - \overline{y}_d x_i)^2$

(D) $s^2_e = \frac{1}{n_d-1}\sum_{i\in S_d}(y_i - \overline{y}_d)^2$

(E) $s^2_e = \frac{1}{n-1}\sum_{i\in S}(u_i - \hat{B})^2$

Q13. In a survey to estimate total agricultural output, a sample of counties is selected via simple random sampling. Within each sampled county, every farm is surveyed. In the terminology of cluster sampling, how are the counties and the individual farms classified?

(A) Counties are primary sampling units (PSUs) and individual farms are secondary sampling units (SSUs).

(B) Counties are secondary sampling units (SSUs) and individual farms are primary sampling units (PSUs).

(C) Counties act as strata, while individual farms act as cluster sampling units.

(D) Both counties and individual farms are considered standalone primary sampling units (PSUs).

(E) Counties represent the sampled population, whereas individual farms represent the target population.

Q14. A school board conducts an SRS of $n = 6$ high schools from a district to estimate the total weekly homework hours (y) using school enrollment (x) as an auxiliary variable. The total district enrollment is known to be $t_x = 12,000$ students. If the sample total enrollment across the 6 schools is $\hat{t}_x = 3,000$ and the overall design weight is $\omega_i = N/n$, what is the adjusted calibration weight ω^*_i for an observation in this sample?

(A) $\omega^*_i = \frac{1}{4}\left(\frac{N}{n}\right)$

(B) $\omega^*_i = 2\left(\frac{N}{n}\right)$

(C) $\omega^*_i = 4\left(\frac{N}{n}\right)$

(D) $\omega^*_i = 6\left(\frac{N}{n}\right)$

(E) $\omega^*_i = 12,000\left(\frac{N}{n}\right)$

Q15. Which of the following provides the most accurate statistical justification for choosing cluster sampling over a simple random sample of individuals, despite cluster sampling typically decreasing estimator precision (increasing variance)?

(A) Cluster sampling guarantees that the sample size within every subpopulation is fixed and predetermined.

(B) Cluster sampling completely eliminates the tracking effect of unobserved shared factors among units.

(C) A complete sampling frame of individual observation units is frequently unavailable, or the logistical and travel costs of an SRS are prohibitively high.

(D) Cluster sampling ensures that the variance of the estimate depends entirely on the variability within clusters rather than between clusters.

(E) It maximizes the architectural heterogeneity of elements chosen across separate primary sampling stages.

Q16. When framing post-stratification as ratio estimation, we show that the post-stratified estimator of the total can be written as a sum of ratio estimators for each post-stratum h, $\hat{t}_{ypost} = \sum_{h=1}^{H} \hat{t}_{0hr}$. To establish this connection using an indicator variable x_{ih} which equals 1 if unit i is in post-stratum h and 0 otherwise, what is the exact value of the known population auxiliary total t_{xh} for post-stratum h?

(A) $t_{xh} = n_h$

(B) $t_{xh} = N_h$

(C) $t_{xh} = N$

(D) $t_{xh} = \frac{N_h}{N}$

(E) $t_{xh} = \sum_{i \in S_h} x_{ih}$

Q17. For the combined ratio estimator under stratified sampling ($\hat{t}_{yrc} = \hat{B}t_x$), the estimated variance depends on the stratum-specific sample residual variances (s_{eh}^2). What is the correct definition of the sample residual variance s_{eh}^2 within stratum h?

(A) $s_{eh}^2 = s_{yh}^2 - 2s_{xyh} + s_{xh}^2$

(B) $s_{eh}^2 = s_{yh}^2 - 2\hat{B}s_{xyh} + \hat{B}^2 s_{xh}^2$

(C) $s_{eh}^2 = s_{yh}^2 - \hat{B}^2 s_{xh}^2$

(D) $s_{eh}^2 = \frac{1}{n_h - 1} \sum_{i \in S_h} (y_{hi} - \overline{y}_{hS})^2$

(E) $s_{eh}^2 = s_{yh}^2 - 2\hat{B}_1 s_{xyh} + \hat{B}_1^2 s_{xh}^2$

Q18. A forestry service needs to choose between stratified sampling and one-stage cluster sampling to estimate total tree biomass across a large conservation area. If the budget restricts data collection to a few concentrated geographic zones, and individual trees within any local area are highly similar to one another due to shared soil and canopy conditions, what is the expected consequence of choosing cluster sampling over stratified sampling?

(A) Cluster sampling will yield a significantly smaller variance than stratified sampling because within-cluster variability is low.

(B) Cluster sampling will yield a larger variance (reduced precision) because the homogeneous clusters provide less new information per sampled unit.

(C) Stratified sampling will be logistically cheaper to implement because it samples from every single zone in the population.

(D) The choice of design will have absolutely no mathematical impact on the precision of the total biomass estimator.

(E) Cluster sampling will completely eliminate the approximate bias of the estimator.

Q19. Consider the textbook's guidance regarding post-stratification variance approximation when the sample size n_h within each post-stratum is reasonably large ($n_h \geq 30$). Under this condition, which sampling design's variance formula serves as a reliable approximation for the post-stratified variance?

(A) Stratified sampling with optimal allocation

(B) Stratified sampling with proportional allocation

(C) Simple random sampling without any weighting adjustments

(D) One-stage cluster sampling with equal probabilities

(E) Regression estimation with a strictly non-zero vertical intercept

Q20. Let y_{ij} represent the measurement for the j-th secondary sampling unit (SSU) within the i-th primary sampling unit (PSU). In the standard population notation for cluster sampling with equal probabilities, which of the following expressions correctly represents the total number of SSUs in the entire population (M_0)?

(A) $M_0 = \sum_{i=1}^{n} m_i$

(B) $M_0 = N \times m_i$

(C) $M_0 = \sum_{i=1}^{N} M_i$

(D) $M_0 = \sum_{i=1}^{N} \sum_{j=1}^{M_i} y_{ij}$

(E) $M_0 = \frac{1}{N} \sum_{i=1}^{N} M_i$

Q21. The performance of a ratio estimator compared to an ordinary SRS sample mean relies heavily on the inequality $R \times \frac{CV(y)}{CV(x)} > \frac{1}{2}$. If a dataset reveals that $CV(y)/CV(x) \approx 1$, what does this relationship imply about the relative variability of the two variables?

(A) Variable x has an absolute standard deviation that is completely identical to the standard deviation of variable y.

(B) Variable y varies similarly relative to its population mean than variable x does relative to its population mean.

(C) The population correlation coefficient R between the two variables is mathematically guaranteed to equal 1.

(D) The absolute variance of the numerator variable is bounded by half the value of the residual variance.

(E) Ratio estimation is guaranteed to be less precise than ordinary simple random sampling.

Q22. The calibration mechanism for a linear regression estimator of a population total is explicitly written as $\hat{t}_{yreg} = \hat{t}_y + \hat{B}_1(t_x - \hat{t}_x)$. Suppose x and y are positively correlated, and a selected sample happens to underestimate the true auxiliary total ($\hat{t}_x < t_x$). What direction of adjustment does this calibration factor force upon the ordinary sample total estimate $\hat{t}_y$?

(A) It adjusts $\hat{t}_y$ downward because the difference $(t_x - \hat{t}_x)$ is negative.

(B) It scales $\hat{t}_y$ by a multiplicative ratio factor matching the proportion $t_x/\hat{t}_x$.

(C) It leaves $\hat{t}_y$ completely unadjusted because regression estimation is strictly unbiased by definition.

(D) It adjusts $\hat{t}_y$ upward because the vertical adjustment factor $\hat{B}_1(t_x - \hat{t}_x)$ adds a positive value.

(E) It forces the final estimate to equal the sample mean square footage parameter.

Q23. When constructing standard confidence intervals for a sub-population domain mean ($\overline{y}_d$), the critical multiplier is selected from a Student's t-distribution. What is the correct number of degrees of freedom used for a domain mean confidence interval under simple random sampling?

(A) $n - 1$

(B) $n - 2$

(C) $n_d - 1$

(D) $n_d - 2$

(E) $N - 1$

Q24. In the standard structural framework of cluster sampling with equal probabilities, let s_r^2 represent the sample variance among estimated primary sampling unit (PSU) totals, defined as $s_r^2 = \frac{1}{n-1}\sum_{i\in S}(\hat{t}_i - \frac{1}{n}\sum_{j\in S}\hat{t}_j)^2$. What structural level of population variability does this metric primarily capture?

(A) The total measurement variability found within individual secondary sampling units.

(B) The relative imbalance between initial design weights and adjusted calibration weights.

(C) The specific structural variability observed between separate primary sampling unit totals in the sample.

(D) The finite population adjustment factor associated with proportional stratum sizes.

(E) The baseline historical biomass divergence across all conservation zones.

Q25. Consider a binary target variable y_i (where $y_i = 1$ if a unit possesses an attribute, and 0 otherwise) evaluated within a specific sub-population domain d. If $\hat{p}_d$ is the sample proportion within domain d, what is the estimated sampling variance $\hat{V}(\overline{y}_d)$ for this domain mean?

(A) $\hat{V}(\overline{y}_d) = (1 - \frac{n}{N})\frac{\hat{p}_d(1-\hat{p}_d)}{n_d}$

(B) $\hat{V}(\overline{y}_d) = (1 - \frac{n}{N})\frac{n}{n_d}\frac{\hat{p}_d(1-\hat{p}_d)}{n-1}$

(C) $\hat{V}(\overline{y}_d) = (1 - \frac{n}{N})\frac{n}{n_d^2}\frac{\hat{p}_d(1-\hat{p}_d)}{n-1}$

(D) $\hat{V}(\overline{y}_d) = (1 - \frac{n}{N})\frac{\hat{p}_d(1-\hat{p}_d)}{n-1}$

(E) $\hat{V}(\overline{y}_d) = (1 - \frac{n}{N})\sum_{h=1}^{H}\frac{N_h}{N}\frac{\hat{p}_d(1-\hat{p}_d)}{n}$

Lecture 7

Q1. In a survey design targeting structural parameters across urban blocks, a researcher assigns initial primary segments as geographic clusters, within which individual housing units are monitored. According to standard survey sampling terminology, how are the blocks and the individual housing units classified?

(A) Blocks are secondary sampling units (SSUs) and housing units are primary sampling units (PSUs).

(B) Blocks are primary sampling units (PSUs) and housing units are secondary sampling units (SSUs).

(C) Blocks serve as strata, whereas housing units function as cluster calibration targets.

(D) Both architectural segments are considered standalone primary sampling units (PSUs).

(E) Blocks represent the sample frame, whereas housing units represent the true observed population.

Q2. A researcher wishes to draw a sample from a multi-tiered administrative structure. Which of the following best describes the core operational difference between a one-stage cluster sampling design and a two-stage cluster sampling design?

(A) One-stage cluster designs stratify the population first, whereas two-stage designs cluster elements iteratively.

(B) One-stage cluster designs require a census of every single secondary sampling unit (SSU) within the selected clusters, whereas two-stage designs sample a subset of SSUs via a secondary simple random sample within those chosen clusters.

(C) One-stage clustering forces the sample size across clusters to be strictly uniform, whereas two-stage designs allow unequal cluster counts.

(D) Two-stage cluster sampling selects primary units with unequal probability, while one-stage models are strictly uniform.

(E) One-stage sampling does not require an overall primary selection phase, jumping directly to sub-unit indexing.

Q3. Let y_{ij} represent the measurement for the j-th secondary sampling unit (SSU) within the i-th primary sampling unit (PSU) under an equal probability design. Which of the following expressions correctly defines the population variance of the primary sampling unit (PSU) totals ($\mathcal{S}_t^2$)?

(A) $\mathcal{S}_t^2 = \frac{1}{n-1}\sum_{i\in S}(t_i - \bar{t}_S)^2$

(B) $\mathcal{S}_t^2 = \frac{1}{N-1}\sum_{i=1}^{N}(t_i - \frac{t}{N})^2$

(C) $\mathcal{S}_t^2 = \sum_{i=1}^{N}\sum_{j=1}^{M_i}\frac{(y_{ij}-\bar{y}_U)^2}{M_0-1}$

(D) $\mathcal{S}_t^2 = \frac{1}{M_0-1}\sum_{i=1}^{N}(t_i - \bar{y}_{iU})^2$

(E) $\mathcal{S}_t^2 = \frac{1}{N-1}\sum_{i=1}^{N}M_i^2(\bar{y}_{iU} - \bar{y}_U)^2$

Q4. In one-stage cluster sampling of equal size ($M_i = M$), a researcher takes a simple random sample without replacement of n clusters out of a total of N population clusters. A common mistake in survey statistics is misidentifying the true number of independent data points obtained. What is the actual number of independent data points used to assess the initial cluster sampling variance?

(A) $n \times M$ data points

(B) $N \times M$ data points

(C) n data points

(D) N data points

(E) $(n \times M) - n$ data points

Q5. Under a one-stage cluster sampling design with equal cluster sizes (M), the population total estimator is constructed as $\hat{t} = N\bar{t}_S$, where $\bar{t}_S = \frac{1}{n}\sum_{i\in S}t_i$. What is the mathematical definition of the sample variance among estimated primary unit totals (s_t^2) used to build the estimated variance of this total?

(A) $s_t^2 = \frac{1}{n-1}\sum_{i\in S}(t_i - \frac{t}{N})^2$

(B) $s_t^2 = \frac{1}{n-1}\sum_{i\in S}(t_i - \bar{t}_S)^2$

(C) $s_t^2 = \frac{1}{M-1}\sum_{j\in S_i}(y_{ij} - \bar{y}_i)^2$

(D) $s_t^2 = \frac{1}{n-1}\sum_{i\in S}M^2(\bar{y}_i - \hat{\bar{y}})^2$

(E) $s_t^2 = \frac{1}{N-1}\sum_{i=1}^{N}(t_i - \bar{t}_S)^2$

Q6. A student wants to estimate the total energy consumption across 100 uniform apartment suites in a complex. He samples $n = 5$ suites at random and monitors every single appliance in those suites (one-stage clustering). The totals for the 5 sample suites are: $t_1 = 13.16$ kWh, $t_2 = 11.36$ kWh, $t_3 = 8.96$ kWh, $t_4 = 12.96$ kWh, and $t_5 = 11.08$ kWh. Calculate the estimated population total $\hat{t}$.

(A) 57.52 kWh

(B) 230.08 kWh

(C) 1150.40 kWh

(D) 575.20 kWh

(E) 28.76 kWh

Q7. A tracking study evaluates student test scores across a population of $N = 100$ uniform suites containing exactly $M = 4$ individuals each. A sample of $n = 5$ suites yields a sample variance among cluster totals of $s_t^2 = 2.88368$. Compute the estimated standard error of the population total estimator, $SE(\hat{t})$.

(A) 13.61

(B) 74.02

(C) 54.79

(D) 165.51

(E) 3.41

Q8. In an architectural context with equal cluster sizes (M), the true population mean estimator per secondary unit is modeled as $\hat{\bar{y}} = \frac{1}{NM}\hat{t}$. What is the exact formula for the true theoretical variance $V(\hat{\bar{y}})$ of this mean estimator under a one-stage cluster sample?

(A) $V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\mathcal{S}_t^2}{nM^2}$

(B) $V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{N^2 \mathcal{S}_t^2}{n}$

(C) $V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n}$

(D) $V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\mathcal{S}^2}{nM}$

(E) $V(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{\mathcal{S}_i^2}{nM^2}$

Q9. When framing a one-stage cluster sampling design using design weights, we define ω_{ij} as the standard inverse inclusion probability for secondary unit j located within cluster i. Under an equal probability one-stage design tracking n sample units from N population clusters, what is the exact weight ω_{ij} assigned to each observed unit?

(A) $\omega_{ij} = \frac{n}{N}$

(B) $\omega_{ij} = \frac{N}{n \times M_i}$

(C) $\omega_{ij} = \frac{N}{n}$

(D) $\omega_{ij} = \frac{M_i}{m_i}$

(E) $\omega_{ij} = \frac{n}{N \times M_0}$

Q10. An analyst coordinates a community infrastructure study across a municipality with $N = 8$ continuous zoning blocks. A simple random sample of $n = 4$ blocks is drawn. Within each sampled block, every single household is successfully tracked (one-stage clustering). If each block contains exactly 5 households, what are the initial sampling weights ω_{ij} for the households included in this dataset?

(A) $\omega_{ij} = 0.5$

(B) $\omega_{ij} = 1.6$

(C) $\omega_{ij} = 2.0$

(D) $\omega_{ij} = 4.0$

(E) $\omega_{ij} = 8.0$

Q11. An investigator implements a one-stage cluster sampling design where primary sampling units (PSUs) are selected with equal probability, and a complete census is conducted within each selected cluster. If the population clusters vary significantly in size ($M_i \neq M$), which of the following best describes the properties of the standard cluster total expansion estimator ($\hat{t} = \frac{N}{n}\sum_{i \in S} t_i$)?

(A) It becomes systematically biased because it fails to incorporate the varying internal cluster weights.

(B) It remains an unbiased estimator of the true population total, but it can exhibit extremely high sampling variance due to the high variability among cluster sizes.

(C) It simplifies mathematically to the stratified proportional allocation total estimator.

(D) Its sampling variance depends exclusively on the within-cluster variance ($\mathcal{S}_i^2$) rather than the between-cluster variance.

(E) It forces the calibration multiplier g_i to converge to exactly 1 across all drawn samples.

Q12. A municipal report on local family earnings presents data gathered from 4 randomly sampled city sectors out of 8 total sectors. Every home within those 4 sectors completed the document. The total sector earnings are discovered to be: $t_1 = 300$, $t_2 = 400$, $t_3 = 208$, and $t_4 = 460$. Calculate the estimated population total $\hat{t}$ utilizing the basic cluster total expansion model.

(A) 1368

(B) 2736

(C) 5472

(D) 342

(E) 10944

Q13. Using the city sector data detailed in the previous question, the sample mean of the cluster totals is calculated as $\bar{t}_S = 342$, and the sample variance among those sector totals evaluates to $s_t^2 = 12,336$. Given that $N = 8$ and $n = 4$, determine the standard error $SE(\hat{t})$ matching the calculations computed by the `survey` package.

(A) 157.08

(B) 314.15

(C) 628.30

(D) 98.69

(E) 444.28

Q14. A researcher analyzes a population of N clusters where the cluster sizes M_i vary significantly from one unit to another ($M_i \neq M$). If a complete census is executed within every selected primary unit, what does m_i (the sample size within cluster i) equal?

(A) $m_i = n$

(B) $m_i = \frac{M_0}{N}$

(C) $m_i = M_i$

(D) $m_i = \overline{M}_S$

(E) $m_i = 1$

Q15. A population contains $N = 100$ variable-sized clusters, and the grand total of all secondary units across the entire frame is known to be $M_0 = 430$. If an investigator draws an SRS of $n = 5$ clusters to calculate a one-stage population mean estimate using the formula $\hat{\bar{y}} = \frac{1}{M_0}\hat{t}$, what is the exact formula for the estimated variance $\hat{V}(\hat{\bar{y}})$?

(A) $\hat{V}(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{s_t^2}{nM_0^2}$

(B) $\hat{V}(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{N^2 s_t^2}{nM_0^2}$

(C) $\hat{V}(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{s_t^2}{n\overline{M}_S^2}$

(D) $\hat{V}(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{ns_t^2}{N^2 M_0^2}$

(E) $\hat{V}(\hat{\bar{y}}) = \left(1 - \frac{n}{N}\right) \frac{M_0^2 s_t^2}{nN^2}$

Q16. Suppose you are running an analysis where the grand total of secondary sampling units in the population (M_0) is completely unknown. To estimate the population mean, you construct a ratio estimator $\hat{\bar{y}}_r = \frac{\bar{t}_S}{\overline{M}_S}$. What baseline parameter does the denominator $\overline{M}_S = \frac{1}{n}\sum_{i \in S} M_i$ track?

(A) The true parameter tracking grand population selection capacity.

(B) The sample mean of cluster sizes observed across the sampled primary units.

(C) The total number of observed units inside a fixed localized multi-stage block.

(D) The expected variance factor corresponding to uniform sample distributions.

(E) The total target response rate calculated via explicit indicator values.

Q17. Under a one-stage cluster design with unequal cluster sizes where the population size M_0 is unknown, an analyst applies the ratio mean estimator $\hat{\bar{y}}_r = \frac{\bar{t}_S}{\overline{M}_S}$. Which of the following defines the correct form of the sample variance s_r^2 used in constructing the ratio standard error?

(A) $s_r^2 = \frac{1}{n-1}\sum_{i \in S}(t_i - \bar{t}_S)^2$

(B) $s_r^2 = \frac{1}{n-1}\sum_{i \in S}(M_i - \overline{M}_S)^2$

(C) $s_r^2 = \frac{1}{n-1}\sum_{i \in S} M_i^2(\bar{y}_i - \hat{\bar{y}}_r)^2$

(D) $s_r^2 = \frac{1}{n-1}\sum_{i \in S}(t_i - \bar{y}_U M_i)^2$

(E) $s_r^2 = \frac{1}{M_0-1}\sum_{i \in S} M_i(\bar{y}_i - \hat{\bar{y}}_r)^2$

Q18. An investigator maps a cluster dataset with variable sizes across $N = 8$ zones, sampling $n = 4$ blocks. The cluster counts for the 4 sampled areas are $M_1 = 5$, $M_2 = 4$, $M_3 = 5$, and $M_4 = 4$. If the sample total biomass is calculated as $\hat{t} = 2426$, compute the estimated total number of units in the population ($\hat{M}_0$) and the resulting ratio mean estimate $\hat{\bar{y}}_r$.

(A) $\hat{M}_0 = 18$, $\hat{\bar{y}}_r = 134.78$

(B) $\hat{M}_0 = 36$, $\hat{\bar{y}}_r = 67.39$

(C) $\hat{M}_0 = 40$, $\hat{\bar{y}}_r = 60.65$

(D) $\hat{M}_0 = 72$, $\hat{\bar{y}}_r = 33.69$

(E) $\hat{M}_0 = 36$, $\hat{\bar{y}}_r = 134.78$

Q19. In a two-stage cluster sampling design tracking unequal cluster dimensions, an SRS of primary segments is performed, followed by a secondary standalone SRS of sub-units inside each selected cluster. What is the expression for the standard design total estimator ($\hat{t}$)?

(A) $\hat{t} = \frac{N}{n}\sum_{i\in S}\overline{y}_i$

(B) $\hat{t} = \frac{N}{n}\sum_{i\in S}M_i\overline{y}_i$

(C) $\hat{t} = M_0 \times \frac{1}{n}\sum_{i\in S}t_i$

(D) $\hat{t} = \sum_{i\in S}\sum_{j\in S_i}\frac{n}{N}y_{ij}$

(E) $\hat{t} = \frac{1}{N\times\overline{M}_S}\sum_{i\in S}t_i$

Q20. The exact theoretical variance of the population total estimator under a two-stage cluster sampling design is modeled using two distinct structural terms: $V(\hat{t}) = N^2(1-\frac{n}{N})\frac{\mathcal{S}_t^2}{n} + \frac{N}{n}\sum_{i=1}^{N}(1-\frac{m_i}{M_i})M_i^2\frac{\mathcal{S}_i^2}{m_i}$. What real-world operational mechanic causes the inclusion of the second component containing $\mathcal{S}_i^2$?

(A) The structural boundary variation observed between global geographical strata.

(B) The deliberate shift toward multi-stage non-response calibration modeling weights.

(C) The sampling variability introduced because we perform a sub-sample (rather than a full census) inside the chosen primary sampling units.

(D) The systemic estimation bias generated when vertical lines fail to target the axis origin.

(E) The standard inflation factor needed to correct for non-probability panel biases.

Q21. When analyzing a two-stage cluster dataset where clusters are selected with equal probability, we state that the tracking weight assigned to a specific observed sub-unit is $\omega_{ij} = \frac{N}{n}\frac{M_i}{m_i}$. Which of the following provides the correct mathematical justification for this structural weight product?

(A) It is the direct sum of the cluster-level and element-level population variance matrices.

(B) It represents the inverse of the joint probability, where $P(\text{unit selection}) = P(\text{cluster selected}) \times P(\text{unit selected} \mid \text{cluster selected})$.

(C) It is a post-stratification coefficient designed to force the target sample mean to equal 1.

(D) It represents an adjustment factor used only when the intraclass correlation is negative.

(E) It maps an optimal multi-stage allocation ratio targeting fixed cost minimization metrics.

Q22. An analyst designs a two-stage cluster survey where primary sampling units (PSUs) are selected with equal probability, and a simple random sample of secondary sampling units (SSUs) is drawn within each selected cluster. If the true population variance within the individual clusters ($\mathcal{S}_i^2$) is remarkably large, which of the following best describes the effect of this internal variability on the precision of the two-stage estimator?

(A) It has absolutely no mathematical impact on the precision because cluster sampling error is completely determined by the variance between cluster totals ($\mathcal{S}_t^2$).

(B) It inflates the second component of the two-stage variance formula, tracking the extra sampling uncertainty introduced because we sub-sample a fraction of elements rather than conducting a full census inside each chosen cluster.

(C) It forces the Intraclass Correlation Coefficient (ICC) to become negative, making the cluster design strictly more efficient than simple random sampling.

(D) It automatically eliminates any requirement to track the primary stage finite population correction factor $(1 - n/N)$.

(E) It alters the initial primary selection stage, converting the design weights from a multi-stage product to a constant uniform inverse probability.

Q23. A two-stage study evaluates property evaluations across 8 city blocks ($N = 8$) by testing 4 blocks ($n = 4$). The underlying cluster sizes are $M_1 = 10, M_2 = 8, M_3 = 11$, and $M_4 = 10$. The secondary sample means for each block are evaluated as $\overline{y}_1 = 60, \overline{y}_2 = 80, \overline{y}_3 = 41.6$, and $\overline{y}_4 = 92$. Compute the overall expansion population total estimate $\hat{t}$.

(A) 2617.6

(B) 5235.2

(C) 1308.8

(D) 7800.0

(E) 10470.4

Q24. When comparing the mathematical precision of an equal-sized one-stage cluster sample against an ordinary simple random sample containing the exact same number of observations ($n \times M$), the textbook derives a variance ratio dependent on the Intraclass Correlation Coefficient (ICC). Which of the following formulas correctly describes this analytical relationship?

(A) $\frac{V(\hat{t}_{\text{Cluster}})}{V(\hat{t}_{\text{SRS}})} = [1 + (M-1)ICC]$

(B) $\frac{V(\hat{t}_{\text{Cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{NM-1}{M(N-1)}[1 + (M-1)ICC]$

(C) $\frac{V(\hat{t}_{\text{Cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{n-1}{N-1}[1 + ICC]$

(D) $\frac{V(\hat{t}_{\text{Cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{M(N-1)}{NM-1}[1 - (M-1)ICC]$

(E) $\frac{V(\hat{t}_{\text{Cluster}})}{V(\hat{t}_{\text{SRS}})} = \frac{1}{M-1}[1 + (M-1)ICC]$

Q25. In real-world survey operations, the Intraclass Correlation Coefficient (ICC) observed within naturally occurring clusters (such as classrooms or neighborhoods) is almost always greater than zero ($ICC > 0$). What does a positive ICC value signify, and what is its statistical effect on the precision of cluster sampling compared to simple random sampling?

(A) It implies that elements inside a cluster are highly diverse, making cluster sampling more precise than SRS.

(B) It implies that elements inside a cluster tend to be highly similar (homogeneous), meaning cluster sampling provides less new information per unit and is less efficient than simple random sampling.

(C) It guarantees that the cluster total estimator will exhibit structural estimation bias.

(D) It forces the finite population correction term at the primary stage to equal 1.

(E) It indicates that variance depends entirely on within-group differences rather than between-group differences.

Answer Key

Lecture 5

Question	Correct Answer	Question	Correct Answer
1	(C) Non-zero Sample Denominator	14	(C) Upward Calibration Shift
2	(B) Ratio Slope Variance Form	15	(A) Approximate Unbiasedness
3	(B) Mean Ratio Calculation	16	(A) Regression Mean Variance Form
4	(B) Overestimation Adjustment	17	(B) Regression Degrees of Freedom
5	(B) Total Biomass Calculation	18	(B) Representative Sample Identity
6	(B) Total Sampling Variance Form	19	(B) Coefficient of Variation Scale
7	(B) Ratio Precision Inequality	20	(A) Closed-bracket Interval Output
8	(C) Mean Asset Value Target	21	(A) Two-sided Divergence Testing
9	(C) Ratio Residual Variance Sum	22	(B) Proportionality Assumption Bias
10	(C) Small Residual Meaning	23	(B) Large Auxiliary Variance Explosion
11	(A) Shared Core Component equation	24	(A) Relative Variability Metric
12	(B) Operational Intercept Rule	25	(B) Tracking Calibration Mechanics
13	(B) Regression Output Mean		

Lecture 6

Question	Correct Answer	Question	Correct Answer
1	(C) Intercept Proportionality	14	(C) Calibrated Weight Scale
2	(D) Variance Calculation	15	(C) Logistical/Frame Justification
3	(B) Ratio Mean Target	16	(B) Post-Stratum Population Size
4	(B) Random Variable Factor	17	(B) Stratum Residual Variance
5	(B) Sample Size Determinism	18	(B) High Cluster Homogeneity Cost
6	(C) Indicator Variable Mask	19	(B) Proportional Proxy Variance
7	(C) Weighted Post-Strat Mean	20	(C) SSU Grand Total Definition
8	(C) Sampling Frame Missing	21	(B) Relative Variation Ratio
9	(B) Proportional Weight Sum	22	(D) Positive Underestimate Shift
10	(C) Homogeneous vs Heterogeneous	23	(C) Active Domain Degrees of Freedom
11	(B) Non-zero Intercept Bias	24	(C) Between-PSU Variance Level
12	(C) Transformed Variable Sum	25	(C) Binary Domain Variance Form
13	(A) PSU and SSU Hierarchy		

Lecture 7

Question	Correct Answer	Question	Correct Answer
1	(B) PSU and SSU Definition	14	(C) Full Census Equivalence
2	(B) Sub-sampling Operational Difference	15	(B) Known Total Variance Form
3	(B) Population Variance of PSU Totals	16	(B) Observed Cluster Mean Size
4	(C) Independent Cluster Counts	17	(C) Ratio Residual Variance Sum
5	(B) Sample Variance of PSU Totals	18	(B) Ratio Domain Size Calculation
6	(C) Global Total Expansion	19	(B) Two-Stage Expansion Total
7	(B) Total Standard Error Factor	20	(C) Sub-sampling Variance Source
8	(A) Uniform Mean Variance Form	21	(B) Multi-stage Joint Probability
9	(C) One-Stage Inclusion Inverse	22	(B) Two-Stage Variance formulation
10	(C) Fixed Design Weight Value	23	(B) Two-Stage Output Calculation
11	(B) Unbiased with high variance properties	24	(B) Analytical ICC Variance Ratio
12	(B) Output Total Expansion	25	(B) Homogeneity Efficiency Loss
13	(B) Sector Standard Error Output		

High level concepts

Ratio vs. Regression Estimators (Lecture 5)

Theoretical Motivation

Simple random sampling (SRS) estimates can exhibit high sampling variance if a randomly selected sample happens to over- or under-represent a key segment of the target population. If a baseline auxiliary variable x is strongly correlated with the target variable y, and its true population total t_x (or mean $\overline{x}_U$) is known from a census or an external register, it can be utilized to calibrate raw sample estimates and substantially increase precision.

Operational Mechanics

- **Ratio Estimation:** Assumes a strictly proportional relationship between the target and auxiliary variables that passes exactly through the origin ($y \approx Bx$). The calibration adjustment factor $g_i = t_x/\hat{t}_x$ dynamically scales individual sample elements depending on whether the sample total of x underestimated or overestimated the true population baseline.
- **Regression Estimation:** Relaxes the strict origin constraint by accommodating a non-zero vertical intercept ($y \approx B_0 + B_1 x$). It applies a dynamic, slope-adjusted additive correction factor $\hat{B}_1(\overline{x}_U - \overline{x}_S)$ to shift the ordinary sample mean baseline.

Variance & Precision

- Ratio estimation yields a smaller mean squared error (MSE) than the unadjusted SRS sample mean when the population correlation tracks the inequality:

$$R \times \frac{CV(y)}{CV(x)} > \frac{1}{2}$$

- The precision of the ratio estimator is measured using the sample residual variance:

$$s_e^2 = \frac{1}{n-1} \sum_{i \in S} (y_i - \hat{B} x_i)^2$$

- Regression estimation captures variability using $s_y^2(1 - r^2)$, making it highly precise whenever a strong linear relationship exists, regardless of the intercept location. It indexes $n-2$ degrees of freedom due to the simultaneous estimation of two separate line parameters $(\hat{B}_0, \hat{B}_1)$.

Domain Estimation vs. Stratified Random Sampling (Lecture 6)

Theoretical Motivation

Investigators routinely need to conduct valid statistical inference on specific sub-populations (such as estimating employment parameters within a specific regional demographic).

Operational Mechanics

- **Stratified Sampling:** Sub-groups (strata) are designated *prior* to drawing the sample, which enables the survey designer to fix and strictly pre-determine the exact sample allocation n_h inside each layer.
- **Domain Estimation:** Sub-groups (domains) are identified *post-hoc* during the statistical analysis phase, typically because domain membership cannot be isolated on the initial sampling frame before selection.

Variance & Precision Adjustments

Because domain membership is unobserved prior to sampling, the active sample size within a target domain (n_d) is a random variable that varies from sample to sample. We notice that n_d is now included in the sampling variance formulas. By framing the domain mean as a specialized ratio estimator utilizing indicator variables ($u_i = y_i x_i$), the estimated sampling variance scales based on the domain sample count:

$$\hat{V}(\overline{y}_d) = \left(1 - \frac{n}{N}\right) \frac{n}{n_d^2} \frac{(n_d - 1)s_{yd}^2}{n - 1}$$

Post-Stratification Frameworks (Lecture 6)

Theoretical Motivation

Post-stratification is executed when a classification variable (e.g., household size) is known to relate strongly to the target variable (e.g., food expenditures), but this metadata is completely missing from the initial sampling frame. Standard stratified sampling is therefore impossible at the design stage.

Operational Mechanics

An overall simple random sample is drawn from the unstratified frame. After data collection, sampled units are classified into their respective post-strata. External census counts ((N_h)) or population proportions ((N_h/N)) are then used to weight the post-stratum sample means:

$$\overline{y} * \text{post} = \sum *h = 1^H \frac{N_h}{N}, \overline{y}_{hS}.$$

To understand the role of these weights, consider what happens if we ignore the external population information and simply compute the overall sample mean, ($\overline{y}_{\mathcal{S}}$). The sample mean can be decomposed by post-stratum as

$$\overline{y}_{\mathcal{S}} = \sum_{h=1}^{H} \frac{n_h}{n}, \overline{y}_{hS},$$

where (n_h) is the realized sample size in post-stratum (h), and (n) is the total sample size.
Similarly, the population mean can be expressed as

$$\overline{y}_{\mathcal{U}} = \sum_{h=1}^{H} \frac{N_h}{N}, \overline{y}_{h\mathcal{U}},$$

where ($\overline{y}_{h\mathcal{U}}$) denotes the population mean within post-stratum (h).

If the sample proportions (n_h/n) differ from the population proportions (N_h/N), the unweighted sample mean places too much emphasis on some post-strata and too little on others. Consequently, it estimates a weighted average corresponding to the sample composition rather than the population composition. Post-stratification addresses this issue by replacing the realized sample proportions (n_h/n) with the known population proportions (N_h/N), thereby aligning the estimator with the target population.

Variance & Precision Adjustments

Just as in domain estimation, individual category sample counts (n_h) are random variables. However, when the sample count within each post-stratum is reasonably large $(n_h \geq 30)$, the variance can be approximated using the standard stratified proportional allocation variance equation:

$$\hat{V}(\overline{y}_{post}) \approx \left(1 - \frac{n}{N}\right) \sum_{h=1}^{H} \frac{N_h}{N} \frac{s_h^2}{n}$$

Ratio Estimation under Stratified Sampling (Lecture 6)

Theoretical Motivation

This design combines the variance-reduction benefits of stratification with the calibration adjustments of auxiliary variables, compounding precision gains.

Operational Mechanics

This is executed using the **Combined Ratio Estimator** $(\hat{t}_{yrc} = \hat{B}t_x)$. A single global slope parameter $\hat{B}$ is computed by dividing the stratified total estimator of y by the stratified total estimator of x:

$$\hat{B} = \frac{\hat{t}_{y,str}}{\hat{t}_{x,str}} = \frac{\sum_{h=1}^{H} N_h \overline{y}_{hS}}{\sum_{h=1}^{H} N_h \overline{x}_{hS}}$$

Cluster Sampling (Lectures 6 & 7)

Theoretical Motivation

Cluster sampling is driven by logistical feasibility and cost containment. A complete sampling frame listing every single individual observation unit is frequently unavailable, or the physical travel costs of sending field interviewers to scattered units under an SRS are prohibitively high.

Operational Mechanics

The population is partitioned into groups known as Primary Sampling Units (PSUs). Individual elements within these clusters are Secondary Sampling Units (SSUs).

- **One-Stage Cluster Sampling:** A random sample of PSUs is selected via an SRS, and a complete census is conducted for *every* SSU inside those chosen clusters $(m_i = M_i)$.
- **Two-Stage Cluster Sampling:** An SRS of PSUs is drawn first, followed by a secondary standalone SRS phase to sub-sample a fraction of the SSUs $(m_i < M_i)$ inside the chosen clusters.

Variance & Precision Adjustments

- **Homogeneity Efficiency Loss:** While stratified sampling increases precision by maximizing differences *between* groups, cluster sampling generally decreases precision (increases variance) compared to an SRS of equal size. Elements within the same cluster naturally tend to be highly homogeneous ($ICC > 0$) due to shared unobserved factors. Sampling multiple repetitive units within the same cluster yields less new information than sampling independent units across the full population.